Exercise 1.6

Required

Textbook p. 34

Let B_i be the event that batch i is selected, and let D denote a defective component. The batch is selected first, so

$$P(B_i) = \frac{1}{4}, \qquad i = 1, 2, 3, 4.$$

The different batch sizes do not change these selection probabilities. Using the stated defect rates,

$$P(D \mid B_1) = 0.05, \quad P(D \mid B_2) = 0.40, \quad P(D \mid B_3) = P(D \mid B_4) = 0.10.$$

(1) Probability of a defective component

By the law of total probability,

$$\begin{aligned} P(D) &= \sum_{i=1}^{4} P(D \mid B_i) P(B_i) \\ &= \frac{1}{4}(0.05 + 0.40 + 0.10 + 0.10) = \boxed{\frac{13}{80} = 0.1625}. \end{aligned}$$

(2) Probability that a defective component came from batch 2

By Bayes' theorem,

$$P(B_2 \mid D) = \frac{P(D \mid B_2) P(B_2)}{P(D)} = \frac{0.40 \times \frac{1}{4}}{0.1625} = \boxed{\frac{8}{13} \approx 0.6154}.$$

The numerical answers use 10% for the two approximately specified defect rates.

Exercise 1.14

Required

Textbook p. 34

(1) Joint distribution function and density

The joint CDF is $F_{XY}(x,y) = P(X \le x, Y \le y)$. Summing the given masses gives

	$y < -1$	$-1 \le y < 0$	$0 \le y < 1$	$y \ge 1$
$x < 0$	0	0	0	0
$0 \le x < 1$	0	0.07	0.25	0.40
$x \ge 1$	0	0.15	0.65	1

Since the variables are discrete, there is no ordinary joint density. In the generalized (Dirac delta) sense requested by the textbook,

$$f_{XY}(x,y) = \delta(x)\big[0.07\delta(y+1) + 0.18\delta(y) + 0.15\delta(y-1)\big] + \delta(x-1)\big[0.08\delta(y+1) + 0.32\delta(y) + 0.20\delta(y-1)\big].$$

(2) Marginal probability mass functions

Summing rows and columns,

$$\boxed{P(X=0) = 0.40, \qquad P(X=1) = 0.60,}$$

$$\boxed{P(Y=-1) = 0.15, \quad P(Y=0) = 0.50, \quad P(Y=1) = 0.35.}$$

All other marginal masses are zero.

(3) Distribution of $Z = XY$

$Z = -1$ only at $(1,-1)$; $Z = 1$ only at $(1,1)$. All remaining outcomes give $Z = 0$. Hence

$$\boxed{P(Z=-1) = 0.08, \quad P(Z=0) = 0.72, \quad P(Z=1) = 0.20.}$$

(4) Correlation coefficient

$$\mathbb{E}X = 0.6, \qquad \mathbb{E}Y = -0.15 + 0.35 = 0.2, \qquad \mathbb{E}(XY) = -0.08 + 0.20 = 0.12.$$

Thus $\mathrm{Cov}(X,Y) = 0.12 - (0.6)(0.2) = 0$. Also,

$$\mathrm{Var}(X) = 0.6 - 0.6^2 = 0.24, \qquad \mathrm{Var}(Y) = 0.15 + 0.35 - 0.2^2 = 0.46.$$

$$\boxed{\rho_{XY} = \frac{0}{\sqrt{0.24 \times 0.46}} = 0.}$$

They are uncorrelated but not independent: $P(X=0, Y=-1) = 0.07 \neq 0.40 \times 0.15$.

Exercise 1.16

Required

Textbook p. 35

Because X and Y are independent standard normal variables,

$$f_{XY}(x,y) = \frac{1}{2\pi}\exp\left[-\frac{x^2+y^2}{2}\right], \qquad x,y \in \mathbb{R}.$$

(1) Joint density of $U = X + Y$ and $V = X - Y$

The inverse transformation is

$$x = \frac{u+v}{2}, \qquad y = \frac{u-v}{2}.$$

Its absolute Jacobian determinant is

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \left|\det\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}\right| = \frac{1}{2}.$$

Since $x^2 + y^2 = (u^2+v^2)/2$, the change-of-variables formula gives

$$\boxed{f_{UV}(u,v) = \frac{1}{4\pi}\exp\left[-\frac{u^2+v^2}{4}\right], \qquad u,v \in \mathbb{R}.}$$

(2) Find the marginal densities and test independence

Integrate the joint density over the other variable:

$$f_U(u) = \int_{-\infty}^{\infty} f_{UV}(u,v)\,dv = \frac{\mathrm{e}^{-u^2/4}}{4\pi}\int_{-\infty}^{\infty}\mathrm{e}^{-v^2/4}\,dv = \frac{1}{2\sqrt{\pi}}\mathrm{e}^{-u^2/4},$$
$$f_V(v) = \int_{-\infty}^{\infty} f_{UV}(u,v)\,du = \frac{\mathrm{e}^{-v^2/4}}{4\pi}\int_{-\infty}^{\infty}\mathrm{e}^{-u^2/4}\,du = \frac{1}{2\sqrt{\pi}}\mathrm{e}^{-v^2/4}.$$

Here, substituting $t = v/2$ gives $\int_{-\infty}^{\infty}\mathrm{e}^{-v^2/4}\,dv = 2\int_{-\infty}^{\infty}\mathrm{e}^{-t^2}\,dt = 2\sqrt{\pi}$. Now compare the product with the joint density:

$$f_U(u)f_V(v) = \frac{1}{4\pi}\mathrm{e}^{-(u^2+v^2)/4} = f_{UV}(u,v) \quad \text{for all } u,v \in \mathbb{R}.$$

Thus U and V are independent. Comparing the marginal densities with $\frac{1}{\sqrt{2\pi\sigma^2}}\exp[-(z-\mu)^2/(2\sigma^2)]$ also gives

$$\boxed{U \sim N(0,2), \qquad V \sim N(0,2), \qquad U \text{ and } V \text{ are independent.}}$$

Exercise 1.21

Required

Textbook p. 35

The given quantities are

$$\mathbb{E}X = 1, \quad \mathbb{E}Y = 3, \quad \mathrm{Var}(X) = 4, \quad \mathrm{Var}(Y) = 16, \quad \rho_{XY} = 0.5.$$

First obtain the covariance:

$$\mathrm{Cov}(X,Y) = \rho_{XY}\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)} = 0.5\sqrt{4 \cdot 16} = 4.$$

Means

Linearity of expectation gives

$$\boxed{\mathbb{E}U = 3\mathbb{E}X + \mathbb{E}Y = 3 + 3 = 6,} \qquad \boxed{\mathbb{E}V = \mathbb{E}X - 2\mathbb{E}Y = 1 - 6 = -5.}$$

Variances

Using $\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X,Y)$,

$$\begin{aligned}\mathrm{Var}(U) &= 9\,\mathrm{Var}(X) + \mathrm{Var}(Y) + 6\,\mathrm{Cov}(X,Y)\\ &= 36 + 16 + 24 = \boxed{76},\end{aligned}$$

$$\begin{aligned}\mathrm{Var}(V) &= \mathrm{Var}(X) + 4\,\mathrm{Var}(Y) - 4\,\mathrm{Cov}(X,Y)\\ &= 4 + 64 - 16 = \boxed{52}.\end{aligned}$$

Covariance

Covariance is linear in each argument, and $\mathrm{Cov}(Y,X) = \mathrm{Cov}(X,Y)$. Thus

$$\begin{aligned}\mathrm{Cov}(U,V) &= \mathrm{Cov}(3X + Y, X - 2Y)\\ &= 3\,\mathrm{Var}(X) - 6\,\mathrm{Cov}(X,Y) + \mathrm{Cov}(Y,X) - 2\,\mathrm{Var}(Y)\\ &= 3(4) - 5(4) - 2(16) = \boxed{-40}.\end{aligned}$$

Here $D(\cdot)$ in the exercise denotes $\mathrm{Var}(\cdot)$.

Exercise 1.24

Required

Textbook p. 35

Assume $a > 0$. We have $X \sim U(0, a)$ and, conditionally on $X = x$,

$$Y \mid X = x \sim U(x, a).$$

(1) Conditional expectation

For $0 \le x < a$, the conditional density is

$$f_{Y|X}(y \mid x) = \begin{cases} \dfrac{1}{a-x}, & x \le y \le a, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore

$$\begin{aligned} \mathbb{E}(Y \mid X = x) &= \int_x^a y \, \frac{1}{a-x} \, dy \\ &= \frac{a^2 - x^2}{2(a-x)} = \frac{x+a}{2}. \end{aligned}$$

Hence

$$\boxed{\mathbb{E}(Y \mid X) = \frac{X+a}{2}.}$$

At the zero-probability endpoint $x = a$, the natural limiting interpretation is $Y = a$, which gives the same formula.

(2) Unconditional expectation

By the law of total expectation and $\mathbb{E}X = a/2$,

$$\begin{aligned} \mathbb{E}Y = \mathbb{E}[\mathbb{E}(Y \mid X)] &= \mathbb{E}\left[\frac{X+a}{2}\right] \\ &= \frac{1}{2}\left(\frac{a}{2} + a\right) = \boxed{\frac{3a}{4}}. \end{aligned}$$

Exercise 1.33

Required

Textbook p. 36

The joint characteristic function is

$$\phi_{XY}(u,v) = \frac{6}{6 - 2\mathrm{j}u - 3\mathrm{j}v - uv}, \qquad \phi_{XY}(u,v) = \mathbb{E}\big[\mathrm{e}^{\mathrm{j}(uX+vY)}\big].$$

(1) Characteristic function of X

Set the argument associated with Y equal to zero:

$$\boxed{\phi_X(u) = \phi_{XY}(u,0) = \frac{6}{6 - 2\mathrm{j}u} = \frac{3}{3 - \mathrm{j}u}.}$$

(2) Expectation and variance of Y

Similarly,

$$\phi_Y(v) = \phi_{XY}(0,v) = \frac{6}{6 - 3\mathrm{j}v} = \frac{2}{2 - \mathrm{j}v}.$$

The transform $2/(2 - \mathrm{j}v)$ corresponds to the density $2\mathrm{e}^{-2y}$ for $y \geq 0$, since $\int_0^\infty 2\mathrm{e}^{-2y}\mathrm{e}^{\mathrm{j}vy}\,dy = 2/(2 - \mathrm{j}v)$. Its first two moments are finite, so differentiating under the expectation gives

$$\phi_Y'(0) = \mathbb{E}[\mathrm{j}Y] = \mathrm{j}\mathbb{E}Y, \qquad \phi_Y''(0) = \mathbb{E}[(\mathrm{j}Y)^2] = -\mathbb{E}[Y^2].$$

Therefore $\mathbb{E}Y = \phi_Y'(0)/\mathrm{j}$ and $\mathbb{E}[Y^2] = -\phi_Y''(0)$. Differentiating,

$$\phi_Y'(v) = \frac{2\mathrm{j}}{(2 - \mathrm{j}v)^2}, \qquad \phi_Y''(v) = \frac{-4}{(2 - \mathrm{j}v)^3}.$$

Thus

$$\phi_Y'(0) = \frac{\mathrm{j}}{2}, \qquad \phi_Y''(0) = -\frac{1}{2},$$

which yields

$$\boxed{\mathbb{E}Y = \frac{1}{2}}, \qquad \mathbb{E}(Y^2) = \frac{1}{2}, \qquad \boxed{\mathrm{Var}(Y) = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4}}.$$

As a check, $\phi_Y(v) = 2/(2 - \mathrm{j}v)$ is the characteristic function of an exponential distribution with rate 2.

Exercise 1.9

Suggested

Textbook p. 34

The probability masses at $1, 2, 3$ are $0.2, 0.5, 0.3$, respectively.

Generalized probability density

A discrete distribution has no ordinary density. Its Dirac delta representation is

$$\boxed{f_X(x) = 0.2\delta(x-1) + 0.5\delta(x-2) + 0.3\delta(x-3).}$$

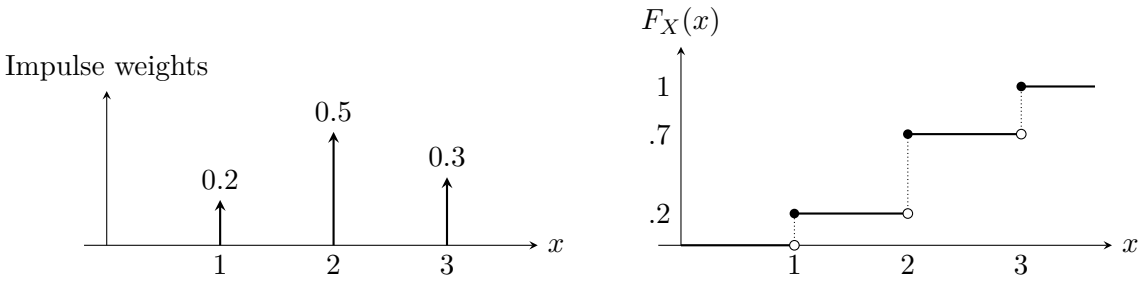
Each impulse weight is a probability mass, rather than a finite density height.

Distribution function

Accumulating the masses gives the right-continuous CDF

$$\boxed{F_X(x) = \begin{cases} 0, & x < 1, \\ 0.2, & 1 \le x < 2, \\ 0.7, & 2 \le x < 3, \\ 1, & x \ge 3. \end{cases}}$$

Graphs



Filled circles indicate included CDF values; open circles indicate excluded endpoints.

Exercise 1.11

Suggested

Textbook p. 34

The density is $f_X(x) = a\mathrm{e}^{-|x|}$ for $x \in \mathbb{R}$.

(1) Normalizing coefficient

A probability density integrates to one. By symmetry,

$$1 = \int_{-\infty}^{\infty} a\mathrm{e}^{-|x|}\,dx = 2a\int_{0}^{\infty} \mathrm{e}^{-x}\,dx = 2a.$$

Therefore

$$\boxed{a = \frac{1}{2}}, \qquad f_X(x) = \frac{1}{2}\mathrm{e}^{-|x|}.$$

(2) Distribution function

If $x < 0$, then $|t| = -t$ throughout the integration interval, so

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{2}\mathrm{e}^{t}\,dt = \frac{1}{2}\mathrm{e}^{x}.$$

If $x \geq 0$, split the integral at zero:

$$\begin{aligned} F_X(x) &= \int_{-\infty}^{0} \frac{1}{2}\mathrm{e}^{t}\,dt + \int_{0}^{x} \frac{1}{2}\mathrm{e}^{-t}\,dt \\ &= \frac{1}{2} + \frac{1}{2}(1 - \mathrm{e}^{-x}) = 1 - \frac{1}{2}\mathrm{e}^{-x}. \end{aligned}$$

Thus

$$\boxed{F_X(x) = \begin{cases} \frac{1}{2}\mathrm{e}^{x}, & x < 0, \\ 1 - \frac{1}{2}\mathrm{e}^{-x}, & x \geq 0. \end{cases}}$$

Both branches give $F_X(0) = 1/2$ by continuity, and the limits at $-\infty$ and $+\infty$ are 0 and 1.

Exercise 1.28

Suggested

Textbook p. 36

Use the convention $\phi_X(v) = \mathbb{E}[\mathrm{e}^{\mathrm{j}vX}]$. Mutual independence implies

$$\begin{aligned}\phi_{b+\sum_{k=1}^{3} a_k X_k}(v) &= \mathbb{E}\left[\mathrm{e}^{\mathrm{j}bv} \prod_{k=1}^{3} \mathrm{e}^{\mathrm{j}va_k X_k}\right] \\ &= \mathrm{e}^{\mathrm{j}bv} \prod_{k=1}^{3} \mathbb{E}[\mathrm{e}^{\mathrm{j}va_k X_k}] = \mathrm{e}^{\mathrm{j}bv} \prod_{k=1}^{3} \phi_k(a_k v).\end{aligned}$$

A coefficient of zero contributes $\phi_k(0) = 1$.

(1) $X = X_1 + X_2$

$$\boxed{\phi_X(v) = \phi_1(v)\phi_2(v).}$$

(2) $X = X_1 + X_2 + X_3$

$$\boxed{\phi_X(v) = \phi_1(v)\phi_2(v)\phi_3(v).}$$

(3) $X = X_1 + 2X_2 + 3X_3$

$$\boxed{\phi_X(v) = \phi_1(v)\phi_2(2v)\phi_3(3v).}$$

(4) $X = 2X_1 + X_2 + 4X_3 + 10$

$$\boxed{\phi_X(v) = \mathrm{e}^{\mathrm{j}10v}\phi_1(2v)\phi_2(v)\phi_3(4v).}$$

Scaling changes the argument of a characteristic function; adding a constant produces a phase factor.

Exercise 1.29

Suggested

Textbook p. 36

Use $\phi(v) = \mathbb{E}[e^{jvX}]$. The mean-square value is $\mathbb{E}[X^2]$, and $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$. Discrete densities below use Dirac deltas.

(1) Sum of five exponential terms

For a discrete variable, $\phi(v) = \sum_k p_k e^{jvx_k}$. Thus the five terms give masses $0.2, 0.3, 0.2, 0.2, 0.1$ at $0, 2, 4, -2, -4$:

$$f_X(x) = 0.2\delta(x) + 0.3\delta(x-2) + 0.2\delta(x-4) + 0.2\delta(x+2) + 0.1\delta(x+4).$$

Consequently,

$$\begin{aligned} \mathbb{E}X &= 0.3(2) + 0.2(4) + 0.2(-2) + 0.1(-4) = \boxed{0.6}, \\ \mathbb{E}[X^2] &= 0.3(4) + 0.2(16) + 0.2(4) + 0.1(16) = \boxed{6.8}, \\ \mathrm{Var}(X) &= 6.8 - 0.6^2 = \boxed{6.44}. \end{aligned}$$

(2) $\phi(v) = 0.3e^{jv} + 0.7e^{-jv}$

$$\boxed{f_X(x) = 0.3\delta(x-1) + 0.7\delta(x+1).}$$

$$\boxed{\mathbb{E}X = 0.3 - 0.7 = -0.4}, \qquad \boxed{\mathbb{E}[X^2] = 1}, \qquad \boxed{\mathrm{Var}(X) = 1 - 0.16 = 0.84}.$$

(3) $\phi(v) = 4/(4 - jv)$

Since $\int_0^\infty 4e^{-4x}e^{jvx}\,dx = 4/(4-jv)$,

$$\boxed{f_X(x) = \begin{cases} 4e^{-4x}, & x \geq 0, \\ 0, & x < 0. \end{cases}}$$

Using $\int_0^\infty x^n e^{-4x}\,dx = n!/4^{n+1}$,

$$\boxed{\mathbb{E}X = \frac{1}{4}}, \qquad \boxed{\mathbb{E}[X^2] = \frac{1}{8}}, \qquad \boxed{\mathrm{Var}(X) = \frac{1}{8} - \frac{1}{16} = \frac{1}{16}}.$$

(4) $\phi(v) = \sin(5v)/(5v)$**, with** $\phi(0) = 1$

The identity $\frac{1}{10}\int_{-5}^{5} e^{jvx}\,dx = \sin(5v)/(5v)$ identifies $X \sim U(-5, 5)$:

$$\boxed{f_X(x) = \begin{cases} \frac{1}{10}, & -5 \leq x \leq 5, \\ 0, & \text{otherwise.} \end{cases}}$$

By symmetry and direct integration,

$$\boxed{\mathbb{E}X = 0}, \qquad \boxed{\mathbb{E}[X^2] = \frac{1}{10}\int_{-5}^{5} x^2\,dx = \frac{25}{3}}, \qquad \boxed{\mathrm{Var}(X) = \frac{25}{3}}.$$

Exercise 1.30

Suggested

Textbook p. 36

Let $X \sim U(a,b)$, where $a < b$. Then

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

Fourier-transform convention

For $\mathcal{F}\{f\}(\omega) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-\mathrm{j}\omega x}\,dx$, the characteristic function satisfies

$$\phi_X(v) = \int_{-\infty}^{\infty} f_X(x)\mathrm{e}^{\mathrm{j}vx}\,dx = \mathcal{F}\{f_X\}(-v).$$

Evaluate the transform

For $v \neq 0$,

$$\begin{aligned} \phi_X(v) &= \frac{1}{b-a}\int_a^b \mathrm{e}^{\mathrm{j}vx}\,dx \\ &= \frac{\mathrm{e}^{\mathrm{j}vb} - \mathrm{e}^{\mathrm{j}va}}{\mathrm{j}v(b-a)} \\ &= \mathrm{e}^{\mathrm{j}v(a+b)/2}\,\frac{\sin(v(b-a)/2)}{v(b-a)/2}. \end{aligned}$$

At $v = 0$, $\phi_X(0) = \int f_X(x)\,dx = 1$, also obtained by taking the limit. Therefore

$$\boxed{\phi_X(v) = \begin{cases} \mathrm{e}^{\mathrm{j}v(a+b)/2}\dfrac{\sin(v(b-a)/2)}{v(b-a)/2}, & v \neq 0, \\ 1, & v = 0. \end{cases}}$$

Common special cases

For $X \sim U(-A, A)$ with $A > 0$,

$$\phi_X(v) = \frac{\sin(Av)}{Av}, \qquad \phi_X(0) = 1.$$

For $X \sim U(0, A)$,

$$\phi_X(v) = \mathrm{e}^{\mathrm{j}vA/2}\frac{\sin(vA/2)}{vA/2}, \qquad \phi_X(0) = 1.$$